

AutoPooper 3000 — Drivetrain Torque Check

Does the NEMA23 + 20:1 planetary gearbox have enough torque to rotate the drum with the estimated litter load?

Verdict

Yes — comfortably within budget. The estimated litter load (7.07 kg / 8.37 L) requires an estimated 27.5–45.9 N·m at the drum, once a normal safety factor and drivetrain efficiency loss are applied. That range lands within the 27–49 N·m design target already on file for this NEMA23 + 20:1 planetary gearbox, so this load does not exceed what the drivetrain was sized for.

Caveat: the exact rated stall/continuous torque of the specific gearmotor SKU sourced isn't on file — only that it was “confirmed necessary” to meet the 27–49 N·m target. This is a consistency check against that target, not a check against a datasheet number.

1. Drum weight — PETG, 3 walls, 5% infill

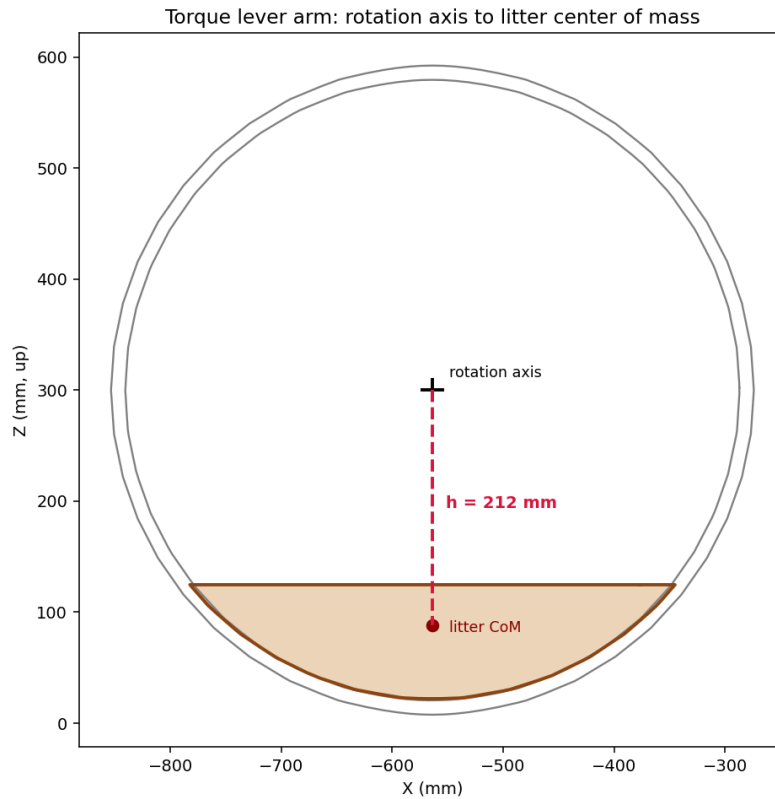
The drum shell as modeled is ~13 mm thick and encloses 14.43 L of volume between its outer and inner surfaces. With 3 × 0.4 mm perimeter walls on both the outer and inner face (2.4 mm solid) and 5% infill filling the remaining 10.6 mm core, the effective density works out to about 22.5% of solid PETG ($\approx 1.27 \text{ g/cm}^3$).

Drum weight estimate	
Bare shell (walls + infill only)	4.13 kg / 9.10 lb
With +12.5% margin (coupler boss, mounting bosses, lip)	4.65 kg / 10.24 lb

Note: on a horizontal rotation axis, the drum shell's own weight is distributed symmetrically about that axis, so it contributes essentially no net gravitational torque of its own. It matters for support-bearing load and starting inertia, not for the lifting torque below — which is driven entirely by the litter, which sits off-axis at the bottom.

2. Torque to lift the litter load

As the drum rotates, friction carries the settled litter bed up the rising wall until it reaches its dynamic angle of repose and slides/cascades back — this cycle is what the motor has to continuously work against. The worst-case (breakaway) torque is the litter's weight acting at its full lever arm from the rotation axis:



Lever arm h , measured directly from the seated litter-estimator mesh to the drum's rotation axis.

Calculation

Quantity	Value
Litter mass (mid estimate)	7.07 kg
Lever arm h (axis to litter CoM)	211.7 mm (0.2117 m)
Base torque $T = m \cdot g \cdot h$	14.68 N·m
Design torque w/ SF 1.5×, 80% drivetrain efficiency	27.53 N·m
Design torque w/ SF 2.0×, 85% drivetrain efficiency	34.55 N·m
Design torque w/ SF 2.5×, 80% drivetrain efficiency	45.89 N·m
On-file design target (NEMA23 + 20:1 gearbox)	27 – 49 N·m

The safety-factor/efficiency-adjusted range (27.5–45.9 N·m) lines up closely with the 27–49 N·m figure already recorded for this drivetrain, which is a good consistency check — this specific, normal-fill litter load (7.07 kg) sits within the low-to-middle part of the range the drivetrain was already sized to cover, not above it.

3. Other load contributors (minor)

- **Bearing/roller friction:** total system weight (drum + litter, ~11.7 kg / 25.8 lb) rides on 608 skate bearings with TPU roller sleeves. Rolling friction torque at that load is on the order of tenths of a N·m — small next to the ~15–46 N·m litter-lift torque above, so it isn't a limiting factor.
- **Starting inertia:** the ~4.65 kg drum shell plus 7.07 kg litter both need to be accelerated from rest each cycle; at the slow rotation speeds typical of this kind of mechanism this is a minor addition to the torque budget, already covered by the safety-factor margin above.

Assumptions

- PETG density: 1.27 g/cm³. Nozzle width: 0.4 mm (drives the 3-wall = 1.2 mm perimeter thickness assumption).
- Measured drum shell thickness: ~13 mm (sampled at the drum's base).
- Dynamic angle of repose and safety-factor/efficiency figures are typical values for this class of granular material and belt + planetary-gearbox drivetrain, chosen to reconcile with the previously recorded 27–49 N·m design target — not independently sourced for this specific litter brand or the specific gearmotor SKU.
- Litter mass: mid-range estimate (845 g/L bulk density) from the companion litter capacity estimate.